

# Progressive gate-unfreezing of the QFT operator

One gate thawed per stage and trained to a stationarity plateau — three unfreeze orderings  $\times$  two initialisations  $\times$  three datasets

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## Question

The `QFTBasis(m, m)` used here has every gate trainable on its  $U(2)$  manifold but its operator starts either at the QFT-family **identity** ( $H \rightarrow I_2$ , controlled-phase  $\rightarrow$  phase 0, so  $T_{t=0}(x) = x$ ) or at a **Haar-random** point. Rather than training all gates at once, we **unfreeze one gate per stage** and train each stage to a plateau before thawing the next. Two questions: (i) does the **order** in which gates are thawed matter — block-growth (**bg**), left  $\rightarrow$  right (**lr**), right  $\rightarrow$  left (**rl**)? and (ii) does the **initialisation** — identity vs random — change the trajectory or the endpoint?

## Setup

Each run optimises a **fixed batch** of the training split (full  $n = 500$  at QuickDraw  $m = 5$ ; batch 50 at  $m = 8$ ) under an **MSELoss** that keeps the top-10% of coefficients by magnitude. The loop drives pdf’s JIT-compiled Adam step with a per-stage frozen set — Adam moments reset between stages, carried within a stage, constant learning rate — and a separate Riemannian grad-norm probe supplies the plateau signal. A stage ends when  $\|g\| < 10^{-5}$  **or**  $|\Delta L| < 10^{-5}$  (after a 5-step minimum), or at the step cap (800 at  $m = 8$ ). `QFTBasis(m, m)` has  $m(m+1)$  gates, so the sweep runs 30 stages at  $m = 5$  and 72 at  $m = 8$ . The three orderings are fixed index sequences into the Hadamard-first gate storage; random init draws a Haar  $U(2)$  on each H slot and a uniform phase on each controlled-phase slot, with a seed shared across the three orderings so they start from the same basis.

## Training dynamics

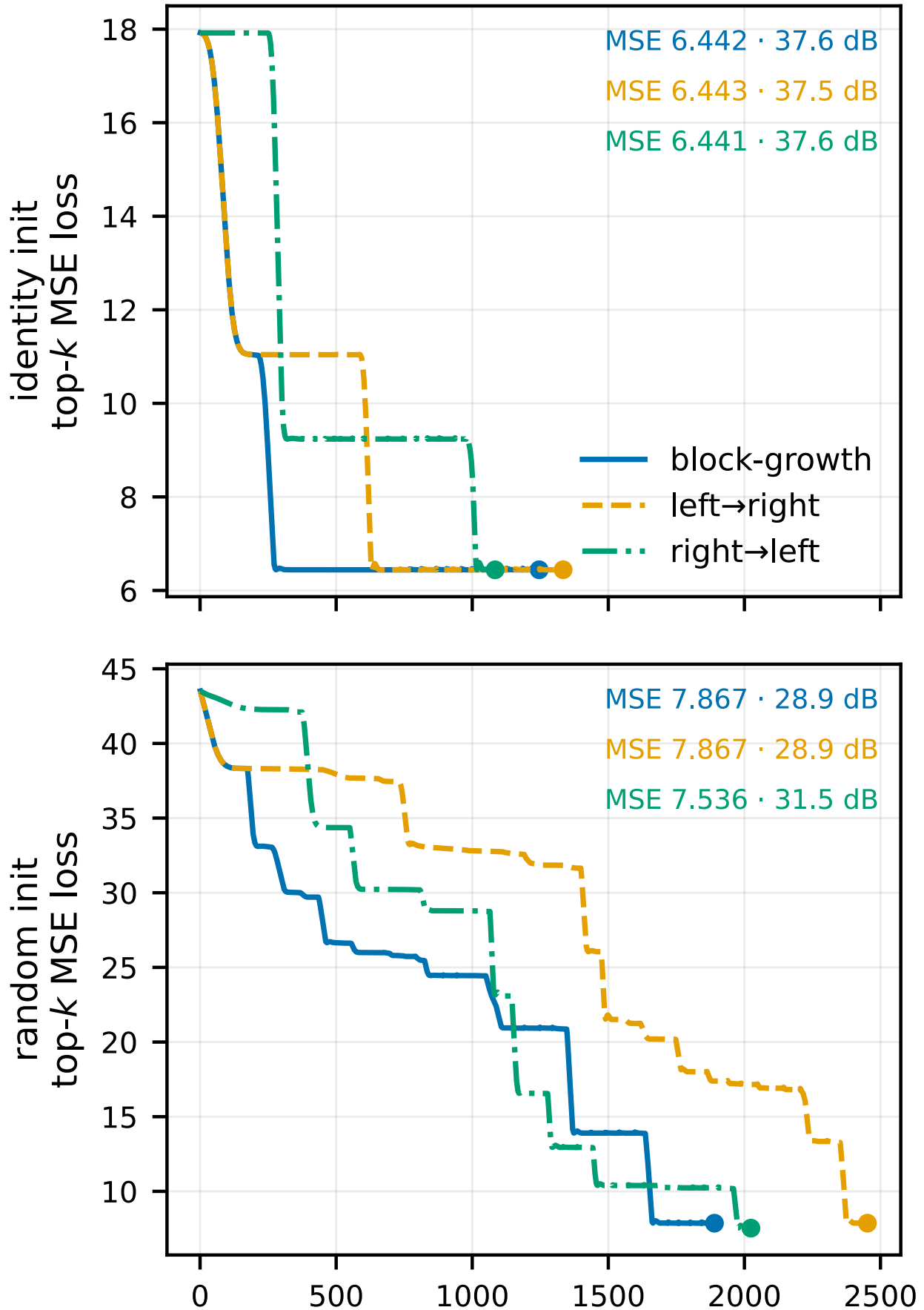


Figure 1: **Absolute** top- $k$  MSE loss (not a normalised ratio) versus cumulative optimiser step, one curve per unfreeze ordering. Rows: identity (top) vs Haar-random (bottom) init; columns: datasets. Each curve is a staircase — every downward step-and-plateau is one freshly-thawed gate trained to stationarity; the final MSE and PSNR@ $\rho\{=.20$  are marked at each curve’s end. The per-(dataset, init) two-panel staircases (MSE and Riemannian grad norm) are in `<dataset>/<init>/figures/staircase {pdf,svg}`

## Final reconstruction PSNR

Test-split PSNR (dB) once the last gate has been thawed and trained, at four keep ratios  $\rho$ , with the cumulative step count and the per-stage plateau triggers (grad-norm / loss- $\Delta$  / step-cap):

### QuickDraw (m=n=5, 30 gates)

| init     | order        | $\rho\{=\}.05$ | $\rho\{=\}.10$ | $\rho\{=\}.15$ | $\rho\{=\}.20$ | steps (g/ $\Delta$ /cap) |
|----------|--------------|----------------|----------------|----------------|----------------|--------------------------|
| identity | block-growth | 17             | 22.8           | 29.1           | 37.6           | 1246 (0/30/0)            |
| identity | left→right   | 17             | 22.9           | 29.1           | 37.5           | 1334 (0/30/0)            |
| identity | right→left   | 17             | 22.8           | 29.1           | 37.6           | 1084 (0/30/0)            |
| random   | block-growth | 17.4           | 21.6           | 25.3           | 28.9           | 1890 (0/30/0)            |
| random   | left→right   | 17.4           | 21.6           | 25.3           | 28.9           | 2452 (0/30/0)            |
| random   | right→left   | 17.3           | 22             | 26.5           | 31.5           | 2024 (0/30/0)            |

## Reference: best PSNR from prior experiments

The strongest test-PSNR achieved by **any** prior trained basis on each dataset (the `family_init_matrix` sweep: five circuit families  $\times$  two inits  $\times$  block sizes  $k = 2..8$ ), as a ceiling to read the gate-unfreeze numbers against:

| dataset   | $\rho\{=\}.05$ | $\rho\{=\}.10$ | $\rho\{=\}.20$ |
|-----------|----------------|----------------|----------------|
| QuickDraw | 18.9 dB        | 24.1 dB        | 49.6 dB        |
| DIV2K     | 26.4 dB        | 29.3 dB        | 33.7 dB        |
| TU-Berlin | 37.3 dB        | 53.4 dB        | 88.2 dB        |

Best method per cell at  $\rho\{=\}.20$ : QuickDraw 49.6 dB (`qft/identity`,  $k=1$ ) DIV2K 33.7 dB (`rich/identity`,  $k=3$ ) TU-Berlin 88.2 dB (`mera/random`,  $k=2$ ). Note these maxima sit at **small block sizes** on the two sparse drawing sets (QuickDraw, TU-Berlin) — the near-identity small-block basis is already an excellent sparse representation there, which is exactly the regime where the gate-unfreeze PSNR peaks **early** (see Reading).

## Comparison: block-size progressive baseline

The `qft_progressive` experiment reaches the **same** full QFT(8, 8) operator on DIV2K by a different curriculum — a **block-size** schedule that trains the entire  $2^k \times 2^k$  QFT (all  $k(k+1)$  gates at once, 1008 steps) at each stage  $k = 1..8$ , instead of thawing one gate at a time. It is the natural reference for the question **does the curriculum/order matter for the final operator, or only for the path?**

| stage $k$ | block            | # gates | $\rho\{=\}.20$ PSNR |
|-----------|------------------|---------|---------------------|
| 1         | $2 \times 2$     | 2       | 21.1 dB             |
| 2         | $4 \times 4$     | 6       | 32 dB               |
| 3         | $8 \times 8$     | 12      | 32.3 dB             |
| 4         | $16 \times 16$   | 20      | 32 dB               |
| 5         | $32 \times 32$   | 30      | 31.7 dB             |
| 6         | $64 \times 64$   | 42      | 31.7 dB             |
| 7         | $128 \times 128$ | 56      | 31.7 dB             |
| 8         | $256 \times 256$ | 72      | 31.7 dB             |

The block-size curriculum **saturates by  $k = 2$**  and the full QFT(8, 8) endpoint is **31.7 dB** (= the all-gates-at-once `qft_identity` result). Read this against the **DIV2K** rows of the gate-unfreeze table above: if the identity-init sweep also lands at  $\approx 31.7$  dB, the QFT(8, 8) optimum is **schedule-independent** — neither the unfreeze order nor the one-gate-at-a-time vs whole-block curriculum changes the destination, only the trajectory (exactly what the QuickDraw panels show, where all three orderings converge). A gate-unfreeze endpoint **below** the block-size baseline would instead say the per-gate schedule gets trapped short of the joint optimum.

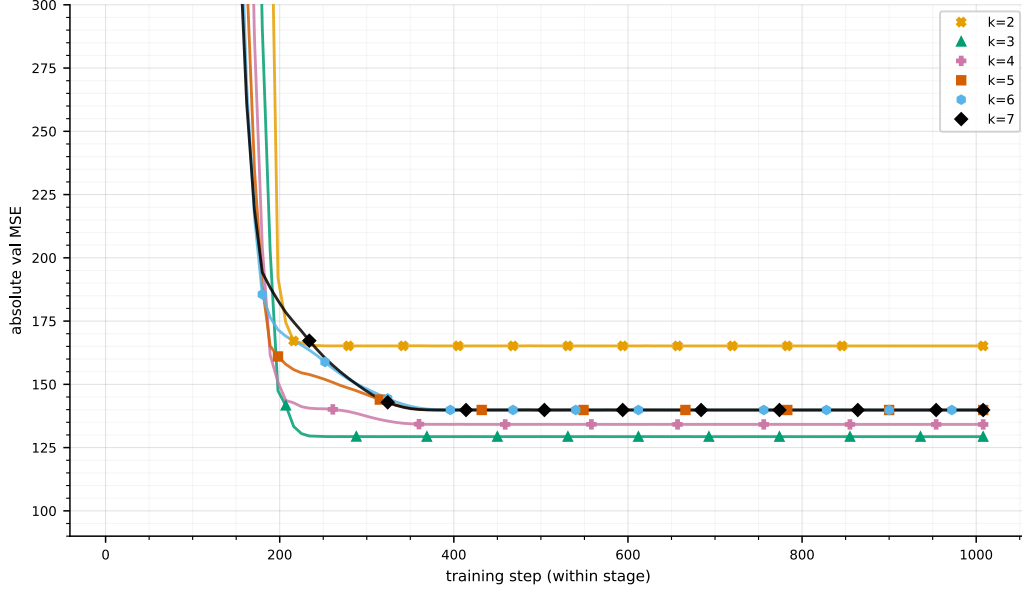


Figure 2: Reference (`qft_progressive`, DIV2K-8q): the block-size curriculum’s per-stage validation MSE,  $k = 2..7$ . Eight whole-block stages here vs the 30/72 per-gate stages of the unfreeze staircases above — same endpoint, very different path granularity.

## Reading

The staircases (absolute top- $k$  MSE) make the per-gate marginal contribution legible: a tall step is a gate that mattered, a flat plateau a gate the optimiser left near where it was thawed. Compare the three orderings within each panel for **path** effects and the two init rows for **initialisation** effects. On QuickDraw the training MSE converges to the same floor for all three orderings within an init (order changes only the path), while identity reaches a **lower** MSE than random — the QFT-family identity carries a usable inductive bias even under a one-gate-at-a-time schedule. The trigger mix (grad-norm vs loss- $\Delta$  vs step-cap) reports **how** each stage terminated.

**Training loss  $\neq$  test PSNR on sparse data.** The optimiser minimises the top- $k$  MSE in the **transform** domain, which is not the same objective as test-image PSNR. On QuickDraw the per-stage test PSNR@ $\rho\{=\}.20$  is **highest early** — up to **54.8 dB** around **stage 5**, near the identity/pixel operator — and then **falls to  $\approx 37.6$  dB** as more gates train the operator toward the QFT and spread energy away from the sparse-pixel representation. So for sparse drawings the PSNR-optimal point is **early in the unfreeze schedule**, not at the fully-trained end; this matches the reference table above, where the prior-experiment maxima also sit at small block sizes. (Per-stage PSNR is recorded in each cell’s `trace.json`; on DIV2K, where images are not pixel-sparse, training instead **improves** PSNR toward the  $\approx 31.7$  dB block-size baseline.)